
Lexical Ontology Persistence and Quantum Models of Meaning: A Theoretical Proposal at the Interface of Semantics, Cognition, and Computation

Abstract

This paper develops a theory of Lexical Ontology Persistence (LOP) and situates it with respect to quantum-inspired models of language meaning. Quantum linguistics and quantum cognition often treat ambiguous words as superpositions of potential senses, with context acting as a measurement that collapses a probabilistic state into a specific interpretation (e.g. Busemeyer & Bruza 2012; Aerts 2009). While this framework captures non-classical contextual effects and graded preferences among readings, it risks conflating **observer-level ambiguity** with the **ontological identity** of lexical items. LOP instead posits that each word possesses an internal, hierarchically structured semantic ontology which persists independently of any particular context, surface form, or momentary interpretation, in line with broader insights from lexical semantics and cognitive linguistics (Lakoff 1987; Evans & Green 2006; Jackendoff 2002).

We articulate LOP in terms of two orthogonal axes: (i) a **vertical axis** capturing the internal ontological hierarchy of a word (historical strata, conceptual cores, argument structure, and usage regimes); and (ii) a **horizontal axis** capturing context-sensitive activation, competition, and collapse among readings. Drawing on examples of polysemy, diachronic semantic layering, and

cross-modal uses of words (in speech, text, images, and digital media) (Traugott & Dasher 2002; Geeraerts 2010), we show how LOP distinguishes stable lexical identity from fluctuating interpretive profiles. We further relate LOP to evidence from psycholinguistics, neurolinguistics, and computational modeling, arguing that persistent lexical ontologies align with distributed yet stable neural patterns (Pulvermüller 1999, 2005) and with structured semantic manifolds in distributional and neural language models (Mikolov et al. 2013; Peters et al. 2018; Devlin et al. 2019). Finally, we propose concrete hypotheses for empirical testing and outline how quantum formalisms can be reinterpreted as working on slices of a deeper lexical ontology rather than as a full theory of word meaning.

1. Introduction

Ambiguous expressions like *bank*, *charge*, or *field* have long motivated probabilistic and non-classical approaches to meaning (Cruse 1986; Tuggy 1993). In quantum linguistics and related quantum cognitive frameworks, an ambiguous word is modeled as a superposition of potential meanings, with context functioning as a measurement that collapses the state into a particular reading (Aerts 2009; Busemeyer & Bruza 2012; Aerts, Gabora & Sozzo 2013). This analogy has been used to capture order effects, incompatibility of questions, and non-classical probability in language understanding and decision making.

However, such models often leave implicit or under-theorized a crucial question: What is the ontological status of the word itself, independent of a given contextual “measurement”? Is *bank* merely a fluctuating cloud of probabilistic potentials, or does it possess an internal semantic architecture that persists across time, media, and contexts? This question echoes long-standing debates in lexical semantics about the internal structure of word meaning (Jackendoff 2002; Pustejovsky 1995; Croft & Cruse 2004).

This paper proposes and develops a theory of Lexical Ontology Persistence (LOP). According to LOP, a lexical item is not just a node in a probabilistic network but a structured semantic

entity with a persistent ontology: historical strata, core conceptual schemas, syntactic and argument-structure properties, and stable relations to other lexical items. Context may activate, suppress, or reweight parts of this ontology, but does not constitute it. This view builds on work in cognitive semantics that treats word meanings as structured networks of related senses grounded in conceptual schemas and metaphor (Lakoff & Johnson 1980; Langacker 1987; Evans & Green 2006).

We argue that quantum formalisms are best understood as modeling **horizontal contextual dynamics** over a pre-existing lexical ontology that unfolds along a **vertical axis**. The vertical axis encodes the internal hierarchical structure and diachronic layering of a word; the horizontal axis encodes momentary contextual configurations, including superposition and collapse. Quantum models, in this view, handle a slice of the full ontology—namely, how interpretations evolve in context—without capturing the ontological hierarchy that LOP foregrounds.

The goals of the paper are threefold:

1. To formulate LOP as a theoretical construct in lexical semantics, building on but going beyond existing accounts of polysemy and lexical networks (Pustejovsky 1995; Cruse 2000; Croft & Cruse 2004).
2. To clarify how LOP relates to, and differs from, quantum-inspired models of meaning (Busemeyer & Bruza 2012; Aerts 2009).
3. To sketch implications for cognitive neuroscience and computational modeling of lexical representations (Pulvermüller 1999, 2005; Mikolov et al. 2013; Peters et al. 2018; Devlin et al. 2019).

The paper is organized as follows. Section 2 reviews relevant work in lexical semantics, quantum linguistics, and cognitive/computational models of word meaning. Section 3 introduces the framework of Lexical Ontology Persistence and the distinction between vertical and horizontal axes of lexical structure. Section 4 illustrates the framework with linguistic data, including polysemy, diachrony, and cross-modal uses. Section 5 compares LOP and quantum formalisms more explicitly. Section 6 relates LOP to evidence from neuroscience and natural language processing and articulates testable predictions. Section 7 concludes.

2. Background

2.1 Lexical semantics and internal structure of words

Classical and cognitive lexical semantics have long recognized that words exhibit **internal structure**, rather than being mere atomic symbols (Lyons 1977; Cruse 1986; Jackendoff 2002). Work in cognitive semantics emphasizes conceptual schemas, frames, and metaphors underlying lexical meaning, where words evoke rich mental representations and structured conceptual networks (Lakoff 1987; Langacker 1987; Evans & Green 2006). On this view, lexical meaning is not simply a list of discrete senses but a web of interrelated sub-meanings organized around core conceptual structures.

Studies of **polysemy** and sense extension show that lexical items maintain a recognizable identity even as they develop multiple related senses (Tyler & Evans 2003; Croft & Cruse 2004; Geeraerts 2010). Radial category models and frame-based accounts assume some central core or structured cluster of senses from which others radiate (Lakoff 1987; Brugman & Lakoff 1988). Diachronic studies likewise show that semantic change tends to preserve traces of earlier meanings, layering new senses over old ones rather than erasing them outright (Traugott & Dasher 2002; Blank & Koch 1999).

From this perspective, a word is not just a label attached to a single stable concept, nor merely a probabilistic distribution over senses, but a **historically and conceptually layered entity**. LOP formalizes this intuition by emphasizing that the internal ontology of a lexical item persists across contexts and through semantic change, echoing the notion of “structured lexical concepts” in Jackendoff (2002) and “qualia structures” in Pustejovsky (1995).

2.2 Quantum models of ambiguity and contextuality

Quantum approaches to cognition and language treat mental states or meanings as vectors in a high-dimensional space, drawing analogies to Hilbert spaces in quantum mechanics (Aerts 2009; Busemeyer & Bruza 2012). In such models, an ambiguous word is represented as a superposition of basis vectors corresponding to distinct senses, and context acts as an operator or measurement that projects this superposition onto a more determinate state. This framework has been applied to explain:

- **Contextuality:** meaning depends on the sequence and nature of contextual cues (Aerts et al. 2013).
- **Incompatibility and order effects:** different question orders and contexts yield non-commutative effects in judgment tasks (Busemeyer & Bruza 2012).
- **Non-classical probability:** violation of classical laws in decision-making and concept combination, including linguistic stimuli (Aerts 2009; Aerts et al. 2013).

These models provide a powerful formal language for how interpretations shift and how probabilities update under context. However, the lexical item itself typically appears as a vector in a state space whose structure is derived from contexts and co-occurrences, rather than from an independently articulated lexical ontology. In other words, the structure of the “meaning space” is often treated as emergent from contextual distributions rather than grounded in a persistent lexical architecture.

2.3 Neural and computational perspectives on lexical representation

In cognitive neuroscience, evidence suggests that word meanings are supported by distributed

neural networks that reflect both modality-specific information (e.g., visual, motor) and more abstract conceptual relations (Pulvermüller 1999, 2005; Huth et al. 2016). Neuroimaging studies show partially overlapping but distinct patterns of activation for different senses of a polysemous word, alongside shared cores that reflect lexical identity (Klepousniotou et al. 2012). Such findings support the idea that lexical meanings are implemented as structured attractor landscapes rather than simple, fixed localist nodes.

In computational linguistics, distributional semantics and neural language models represent words as vectors or contextualized embeddings learned from large corpora (Turney & Pantel 2010; Mikolov et al. 2013; Pennington et al. 2014; Peters et al. 2018; Devlin et al. 2019). These representations capture rich statistical structure and often reveal clustering corresponding to semantic relations, polysemy, and syntactic roles (Reisinger & Mooney 2010; Neelakantan et al. 2014). At the same time, the persistence of a token's representation across diverse contexts suggests some form of internal structure: the embedding space behaves like a semantic manifold within which context traces trajectories (Ethayarajh 2019).

LOP can be seen as a theoretical bridge linking these perspectives: the persistent lexical ontology corresponds to structured regions of neural and embedding space, while contextual dynamics correspond to trajectories through and projections onto those regions.

3. Theoretical Framework: Lexical Ontology

Persistence

3.1 Core idea

Lexical Ontology Persistence (LOP) is the claim that for any lexical item w , there exists a

persistent internal ontology—a structured, multi-layered semantic object—that:

- Exists independently of any particular token instance, context, or observer (in line with structuralist and generative conceptions of lexical entries; cf. Jackendoff 2002; Pustejovsky 1995).
- Persists diachronically, even as new senses are added and usage conditions change (Traugott & Dasher 2002).
- Organizes the word's possible uses along a hierarchical semantic architecture, rather than merely as a flat set of senses with associated probabilities (Cruse 2000; Geeraerts 2010).

On an intuitive level, LOP says: the word *bank* has a stable internal ontology that includes, among other things, financial institutions, river edges, and related conceptual and syntactic structures. Context does not create these options; it selects, weights, or folds within a pre-given ontology. This is consistent with cognitive semantic accounts of structured polysemy and radial categories (Lakoff 1987; Tyler & Evans 2003), but LOP emphasizes the metaphysical persistence of the ontology irrespective of particular contextual instantiations.

3.2 Vertical vs. horizontal axes

We can visualize LOP using two orthogonal axes, borrowing the intuition of manifolds from both cognitive semantics and geometric approaches in NLP (Turney & Pantel 2010; Ethayarajh 2019):

- The vertical axis (Y) represents the ontological hierarchy of the word:
 - Historical layers of meaning (Traugott & Dasher 2002).
 - Core conceptual schemas and metaphors (Lakoff 1987; Evans & Green 2006).
 - Argument structure and selectional restrictions (Jackendoff 2002; Levin & Rappaport Hovav 2005).
 - Relations to other lexical items (synonymy, antonymy, derivation; Cruse 1986).
- The horizontal axis (X) represents contextual dynamics:

- Activation of senses in a given discourse (Klein & Murphy 2001).
- Competition, superposition, and collapse among candidate meanings, modeled via probabilistic or quantum formalisms (Aerts 2009; Busemeyer & Bruza 2012).
- Pragmatic and stylistic modulations (Sperber & Wilson 1995).
- Media-specific realizations (spoken conversation, legal text, meme image, caption, etc.).

Quantum-inspired models primarily operate along X: they provide tools for modeling how a system moves between states under contextual operations. LOP asserts that the full nature of lexical meaning requires modeling Y as well: the vertical, ontological axis that contextual dynamics can only traverse, not define.

3.3 Ontological vs. epistemic ambiguity

Within LOP, we distinguish:

- **Ontological ambiguity:** genuine multiplicity in the internal ontology (e.g., homonymy, deeply divergent senses, structured polysemy; Cruse 1986; Croft & Cruse 2004).
- **Epistemic ambiguity:** uncertainty on the part of the interpreter given limited contextual information (e.g., in processing ambiguous sentences; see Klein & Murphy 2001).

Quantum superposition can be interpreted as a mixture of both: the vector encodes ontological possibilities and epistemic uncertainty (Aerts 2009; Busemeyer & Bruza 2012). LOP insists that the ontological component is **not reducible** to the epistemic one: even if an interpreter had complete information, the word would still possess a layered ontology, because its history and conceptual structure are inherently multi-faceted (Traugott & Dasher 2002; Geeraerts 2010).

4. Data and Phenomena

To illustrate LOP, we consider three classes of phenomena: polysemy and homonymy, diachronic layering, and cross-modal “slices” of reality.

4.1 Polysemy and homonymy

Consider the English word *charge*:

1. *The battery is out of charge.*
2. *They will charge you a fee.*
3. *The knight began to charge.*
4. *The particles carry opposite charges.*

A quantum model may represent *charge* as a superposition of senses—electrical, financial, motion, legal, etc.—with context determining which basis vector is selected or how amplitudes change (Aerts 2009; Busemeyer & Bruza 2012).

LOP instead starts by asking: What is the internal ontology of ***charge***? We can distinguish:

- A cluster relating to energy and potential (physical and metaphorical).
- A cluster relating to financial obligations and transactions.
- A cluster relating to responsibility and accusation.
- A cluster relating to forceful motion or attack.

These clusters are not arbitrary; they reflect historical derivation, metaphorical extension, and systematic syntactic patterns—precisely the kinds of regularities studied in work on lexical rules and qualia structures (Pustejovsky 1995; Levin & Rappaport Hovav 2005). The internal ontology of *charge* organizes these clusters in a hierarchical structure, with paths connecting literal physical meanings to metaphorical and institutional ones (Lakoff & Johnson 1980; Evans & Green 2006). Quantum dynamics over this ontology can model which cluster becomes active in context, but cannot by itself explain why these clusters exist and are related as they are.

4.2 Diachronic semantic layering

Diachronic change shows that words typically add layers rather than replacing their ontologies wholesale. For instance, many verbs of physical motion acquire abstract or metaphorical senses related to social status, information flow, or emotional states (Sweetser 1990; Traugott & Dasher 2002). The original physical schema persists as an underlying structure, even if some senses become less frequent. Historical corpora and semantic change studies repeatedly show how new meanings emerge through processes such as metaphor, metonymy, and subjectification while retaining traces of earlier uses (Blank & Koch 1999; Traugott & Dasher 2002).

From an LOP perspective, diachronic change is the growth and reconfiguration of the vertical axis. New branches are added to the ontological tree; old branches become thinner but are rarely deleted entirely. This continuity explains why speakers can still exploit older senses in poetic or playful usage and why diachronic traces often remain detectable in synchronic patterns (Geeraerts 2010).

A purely quantum model that privileges current co-occurrence statistics risks treating older or rare senses as negligible low-amplitude components, thereby underrepresenting the historical depth of the ontology. LOP insists that even low-frequency branches are part of the persistent lexical object.

4.3 Cross-modal slices of reality

Words are used in different slices of reality: spoken interaction, written text, formal documents, casual chat, social media posts, labels on images, captions in videos, and interactions with AI systems. Each medium imposes different constraints and affords different contextual cues (Coupland 2007; Androutsopoulos 2011).

Consider a word like *field*:

- In a physics article (*field theory*).
- In an agricultural context (*wheat field*).
- In a user interface (*field* in a form).
- In a sports broadcast (*on the field*).

Each medium—scientific paper, farming report, UI documentation, sports commentary—activates a different slice of the word’s ontology. LOP treats these as horizontal projections of a single persistent vertical ontology that ties together the conceptual core (spatial extension, region, domain of interaction) with its institutionalized uses in different domains (Cruse 1986; Geeraerts 2010).

Quantum contextual dynamics can model how identical surface forms (*field*) behave differently in these media (Aerts et al. 2013), but it is LOP that explains why these uses are still recognizably related and why speakers can move between them with deliberate metaphor and wordplay.

5. LOP and Quantum Formalisms

5.1 Contextual collapse as projection on a lexical

manifold

In a quantum-inspired view, meanings live in a vector space, and context acts as a projection operator (Busemeyer & Bruza 2012; Aerts 2009). LOP suggests that we interpret this vector space not as an undifferentiated cloud of potentials but as a **structured manifold** shaped by the lexical ontology. The space is not arbitrary; it is *curved* and *folded* by the word's internal structure, much as conceptual spaces proposed in cognitive science are structured by quality dimensions and prototypes (Gärdenfors 2000).

Contextual “collapse” is then a projection not onto a random axis but onto specific **submanifolds** corresponding to branches of the ontology (e.g., financial vs physical senses). LOP thereby constrains the quantum model: not all superpositions are equal; some are licensed and supported by the ontological hierarchy, while others are structurally disfavored or incoherent.

5.2 Non-classical probability and lexical competition

Quantum probabilities can capture interference effects and non-classical phenomena in meaning selection (Aerts 2009; Busemeyer & Bruza 2012). Within LOP, these effects arise **horizontally**: competition between branches of the ontology under conflicting or underspecified contextual cues.

For example, a sentence like *The bank was flooded* involves competition between **financial institution** and **river side** senses. Contextual cues (e.g., surrounding words like *river*, *water*, *customers*) modulate amplitudes, akin to priming and cue-based retrieval effects observed in psycholinguistics (Klein & Murphy 2001). Interference may arise when cues support overlapping but not identical subregions of the ontology. LOP does not replace this dynamic; it provides a structured space in which such dynamics occur.

5.3 Limits of purely quantum approaches

Purely quantum approaches, if they ignore LOP, face at least three limitations:

1. **Ontological opacity:** They encode which senses are likely but not *why* these senses exist or how they historically and conceptually relate (cf. Traugott & Dasher 2002; Geeraerts 2010).
2. **Diachronic blindness:** They typically treat current distributions as sufficient, underplaying the structured persistence of older senses (Blank & Koch 1999).
3. **Media insensitivity at the ontological level:** They can model media-specific probabilities, but not how different media carve out systematic slices of a single underlying ontology (Coupland 2007; Androutsopoulos 2011).

LOP therefore does not compete with quantum linguistics as a formalism for contextual dynamics; rather, it reframes it as a **horizontal theory** operating on a **vertical ontological substrate**.

6. Implications for Cognitive Neuroscience and Computational Modeling

6.1 Neural implementation of persistent ontologies

If LOP is correct, we expect neural representations of word meaning to show:

- **Stable yet distributed cores:** shared activation patterns across contexts that reflect the word's persistent ontology (Pulvermüller 1999, 2005; Huth et al. 2016).

- **Branch-specific modulations:** additional activity patterns associated with specific senses or branches (e.g., financial vs physical *charge*), as suggested by studies on polysemy and homonymy in the brain (Klepousniotou et al. 2012).
- **Diachronic echoes:** partial reactivation of historically earlier senses in metaphor, wordplay, or archaic registers, consistent with the persistence of older conceptual structures (Traugott & Dasher 2002).

Such patterns would support the idea that the brain encodes lexical items as **structured attractor landscapes**, where context-driven trajectories traverse a relatively stable manifold rather than re-creating meaning from scratch each time (Pulvermüller 1999).

6.2 NLP models as approximations of lexical manifolds

Modern large language models can be interpreted as learning an approximation of lexical ontologies and their contextual dynamics. Under LOP, we can interpret:

- **Static embeddings** (e.g., word2vec, GloVe) as coarse projections of the vertical ontology (Mikolov et al. 2013; Pennington et al. 2014).
- **Contextualized embeddings** (ELMo, BERT, etc.) as trajectories along the horizontal axis for particular token instances (Peters et al. 2018; Devlin et al. 2019; Ethayarajh 2019).
- **Sense clustering and subspace methods** as attempts to recover branches of the underlying lexical ontology (Reisinger & Mooney 2010; Neelakantan et al. 2014).

Quantum-inspired architectures in NLP—e.g., models that use density matrices or complex-valued representations—can be reinterpreted as providing richer **horizontal operators** on an embedding space that is implicitly shaped by LOP-like constraints, even if these are not explicitly modeled (e.g. Bruza et al. 2009).

6.3 Empirical predictions

LOP yields several testable predictions:

1. **Cross-task stability:** For a given word, patterns of neural or embedding similarity among senses will preserve a hierarchical structure across tasks and contexts, consistent with a persistent ontology (Klepousniotou et al. 2012; Ethayarajh 2019).
2. **Diachronic recoverability:** Historical senses that are rare in contemporary corpora should still be recoverable as coherent subclusters in large-scale embeddings (Hamilton et al. 2016), reflecting the residual structure of the vertical axis.
3. **Media-slice coherence:** Uses of a word across very different media (speech, formal writing, memes) will map onto overlapping subregions of a single manifold rather than forming entirely disjoint spaces (Coupland 2007; Androutsopoulos 2011).

Experimental and corpus-based work can test these predictions by combining diachronic corpora, neuroimaging, and probing analyses of neural language models.

7. Conclusion

This paper has proposed Lexical Ontology Persistence (LOP) as a theoretical framework for understanding the nature of word meaning. LOP asserts that lexical items possess persistent internal ontologies—hierarchically structured, diachronically layered semantic architectures that exist independently of any particular context or observer (Cruse 1986; Jackendoff 2002; Traugott & Dasher 2002). We argued that quantum-inspired models of meaning are powerful tools for describing horizontal contextual dynamics—superposition, interference, collapse—but that they operate on slices of a deeper vertical ontological axis that they do not fully capture (Aerts 2009; Busemeyer & Bruza 2012).

By distinguishing ontological from epistemic ambiguity and by modeling words in terms of vertical ontologies and horizontal contextual slices, LOP offers a way to integrate insights from lexical semantics, diachronic linguistics, psycholinguistics, neurolinguistics, and computational modeling (Pulvermüller 1999; Geeraerts 2010; Mikolov et al. 2013). It suggests that persistent lexical ontologies correspond to structured manifolds in both neural and computational representational spaces, while quantum formalisms describe how context moves within and projects onto those manifolds.

Future work should formalize LOP more explicitly, develop quantitative measures of ontological persistence, and test its predictions using behavioral, neuroimaging, and large-scale corpus methods. In doing so, we may arrive at a more complete theory in which quantum linguistics and lexical ontology persistence are not rivals but complementary components of a unified account of meaning.

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